Long-term analysts' forecasts and real earnings management

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Abstract: This paper investigates the implications of earnings manipulations through real activities for financial analysts. In particular, it examines the extent to which analysts consider the effects of real earnings management on company performance when they make long-term earnings growth projections. Our results show analysts issue lower long-term earnings growth forecasts for firms with higher levels of real earnings management. In addition, we find analysts produce less accurate long-term earnings forecasts in the presence of real activities manipulation, and that the forecast bias increases with the level of real earnings management. Our results from the sensitivity tests confirm our findings that financial analysts are able to recognise real earnings management and penalise firms that engage in it when they issue long-term earnings growth forecasts.

Keywords: long-term analysts' forecasts; analysts forecasts; earnings growth forecasts; earnings management; real earnings management.

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1 Introduction

This study examines the implications of real earnings management for long-term analysts' forecasts. Researchers have identified several management mechanisms executives use to affect earnings numbers, including real activities earnings management (REM), accrual earnings management (AEM), and classification shifting of income statement items. Research investigating REM has gained more traction since the appearance of empirical evidence suggesting that REM is more prevalent, especially after the passage of the Sarbanes-Oxley (SOX) act of 2002. Graham et al. (2005) note that approximately 80% of executives prefer to decrease discretionary expenses, including maintenance, research and development, and advertising, to reach an earnings target. They rationalise this preference by claiming that auditors are lesser prone to challenge real economic actions than accounting accruals. Other research shows that firms have increasingly shifted from AEM to REM in the post-SOX era (e.g., Ewert and Wagenhofer, 2005; Cohen et al., 2008; Zang, 2012; Chan et al., 2015).

Most studies on REM (e.g., Baber et al., 1991; Bartov, 1993; Bushee, 1998; Herrmann et al., 2003; Roychowdhury, 2006; Cohen et al., 2010; Sun, 2021) investigate

how executives make managerial decisions specifically to meet or exceed specific earnings targets. As opposed to AEM, REM involves a deliberate alteration of the operational decisions to inflate performance over the short term artificially. Altering operational decisions for short-term gain may have a detrimental effect on the company's future performance. For example, extant research suggests that REM may cause real economic costs (Ewert and Wagenhofer, 2005) and can adversely affect operating performance over time (Cohen and Zarowin, 2010; Eldenburg et al., 2011; Vorst, 2016). This detrimental effect resulting from making sub-optimal operational decisions may persist in the long run and affect investors' and other market participants' decision-making processes. It is, therefore, essential to understand whether investors are aware of firms' REM practices both in the short and long term. Given that investors' decisions, by and large, rely on analysts' assessments and predictions, we propose to investigate if analysts detect REM and whether they incorporate this information when they issue long-term earnings growth forecasts.

Studies investigating analysts' reactions to real earnings management are crucial since analysts play a vital role in stock price setting and since any changes in their perceptions are likely to affect their forecasts. Both Yu (2008) and Hong et al. (2014) suggest that financial analysts play a monitoring role by inhibiting management from engaging in REM and AEM. They add that analyst following and coverage are associated with less earnings management. More recently, two studies, Lee and Chung (2019) and Eiler et al. (2021), directly investigate whether security analysts incorporate firms' REM in their forecasts. The collective empirical evidence from these studies suggests analysts identify and consider whether firms engage in REM and reflect that awareness through their earnings forecasts and stock recommendations. Both studies, however, limit their analysis to short-term horizon earnings forecasts. We diverge from these studies by investigating the effect of REM on long-term forecasts.

Long-term analyst forecasts reflect analysts' anticipation regarding firms' long-term earnings growth. A well-established body of research shows that long-term forecasts are incrementally useful to investors in estimating firm value above and beyond short-term forecasts (e.g., Brown et al., 1985; Mest and Plummer, 1999; Bandyopadhyay et al., 1995, Srinidhi et al., 2009; Da and Warachka, 2011; Jung et al., 2012). Although prior literature suggests that REM has implications for short-term analysts' forecasts and that long-term forecasts are intrinsically valuable to investors, there is no empirical evidence on whether REM affects long-term analyst forecasts. Contrarily, it is plausible that analysts may consider that REM effects dissipate over time and thus are irrelevant when issuing long-term forecasts. In addition, for some manipulation tactics, such as the case of research and development activity manipulation, executives may manage earnings to smooth the income and improve earnings informativeness (Baik et al., 2022). This study provides empirical evidence regarding these competing theories. It remains unclear whether analysts detect earnings manipulation through real activities and whether they reflect this knowledge in their long-term forecasts. In this research, we aim to address this question and fill this void in the literature.

Using a sample of US firms from 2002 to 2017, we first investigate whether REM proxies are associated with analysts' long-term earnings growth forecast. We also examine whether REM variables are associated with long-term earnings growth forecast errors. We find that analysts issue lower long-term earnings growth forecasts for corporations with greater REM levels. In addition, we find that analysts' long-term earnings forecast bias increases with the level of REM. Overall, our results indicate that

analysts are able to detect REM activities and reflect their discovery in their long-term earnings growth forecasts.

This study adds to the growing REM body of research. We contribute to this line of inquiry by examining the implications of REM on long-term earnings forecasts. Despite the documented evidence of an adverse effect of REM on firms' operating performance over the long run (e.g., Vorst, 2016), no prior research investigates the association between REM and long-term analyst forecasts. This study is the first to provide empirical evidence of this association. This paper extends both Lee and Chung (2019) and Eiler et al. (2021) by examining the extent to which analysts integrate the effects of REM on future performance when they issue long-term earnings growth forecasts. Finally, our research contributes to the analyst forecast literature by considering REM as a source of analyst forecast bias and investigating its consequences on analyst forecasts over the long run.

In the present study, we focus on the analysts' views of REM. In particular, we examine whether financial analysts detect and consider when corporations take part in REM activities and whether they incorporate this information in their earnings growth forecasts. This question is specifically important, considering the recent findings presented in Espahbodi et al. (2022) regarding the long-term persistence of REM activities. While the authors find that REM levels have decreased over time, especially after the passage of relevant regulations, they note that REM continues to persist in the long-term post-regulation periods. Our findings are significant for various market players because they suggest that, besides the short-term, financial analysts are able to detect and impound their discovery of REM in the long-term earnings growth forecasts. Managers who think that REM is hard to uncover and that they may go undetected when they engage in REM should be wary. While these managers may believe that REM may not have implications for the long run, our conclusions should caution them that analysts can uncover their manipulation tactics and factor in their discovery when issuing long-term forecasts. In addition, investors can be more confident that analysts recognise the adverse impact of REM on firms' long-term performance and will signal their discovery in long-term forecasts. Our results are also relevant for auditors and regulators. Since long-term forecasts are essential to the smooth functioning of financial markets and the economy (Srinidhi et al., 2009), our findings warrant the need for effective strategies to deter, detect, and address the REM tactics plaguing financial markets.

We organised the paper as follows. We first review the current literature and propose the hypotheses. We then explain the research design and present the sample data. Section 4 discusses the findings from the univariate and multivariate analyses. Section 5 presents the conclusion.

2 Background and related work

Earnings management has been quite popular topic in the accounting field and continues to be grounds for further research. Although previous research lent considerable attention to AEM, more recently, there has been an increasing interest in real earnings management (REM). In fact, according to Graham et al. (2005), executives have expressed more interest in manipulating earnings using REM than AEM. In addition, choices to manipulate earnings via corporate decisions that are considered 'real' are given priority over choices to manipulate them via accruals (Zhang, 2012).

It is common for corporations to change their business practices purposefully to attain performance objectives. Extant literature enumerates the following REM tactics: overproduction, fixed asset sales manipulation, discretionary expense manipulation, and regular sales manipulation. It documents how executives use these tactics to meet or exceed earnings benchmarks.

The overproduction manipulation tactic is used by manufacturing firms and relates to the executives' efforts to decrease the cost of goods sold by manufacturing more product quantities. Doing so spreads the fixed overhead cost over more units and thus increases earnings. Empirical evidence suggests that manufacturing companies use overproduction to decrease the current cost of goods sold, thus boosting current earnings (e.g., Thomas and Zhang, 2002; Roychowdhury, 2006).

The fixed asset sales manipulation consists in strategically 'timing' the sale of an asset so that the recognition of income from the asset disposal falls in the desired period. The fixed asset sales tactic is effective when an asset's market value exceeds its book value and a gain from the disposal of the asset is realised. Although the gain from the sale must be recorded in the disposal period, the sale's timing remains the executive's prerogative. Prior literature suggests that executives avoid negative earnings performance by manipulating the timing of long-term asset sales (Bartov, 1993). Other corroborating evidence suggests that executives use gains from the disposal of long-term assets to manage earnings and that firms display earnings increases (decreases) via fixed asset sales when current earnings dip below (above) management forecasts (Hermann et al., 2003).

Discretionary expense manipulation refers to the executives' use of their discretion to reduce expenses such as selling, general and administrative (SG&A), research and development (R&D), advertising, and other discretionary expenses, to boost earnings in the current period. For example, firms cut their R&D expenses significantly when such spending threatens the achievement of earnings benchmarks (Bushee, 1998; Baber et al., 1991). Similarly, managers cut advertising spending to avoid earnings decreases and losses. For instance, firms tend to decrease advertising budgets in a fiscal quarter's third month and in the last quarter to exceed earnings from the preceding year (Cohen et al., 2010). Anecdotally, 80% of executives report they would choose to decrease discretionary spending to achieve earnings objectives Graham et al. (2005).

Another method of REM is through regular sales manipulation. This manipulation avenue refers to temporarily shifting sales from the upcoming periods into the current period by offering more attractive sales terms and discounts, for instance. Assuming that sales margins are positive, temporarily boosting sales volume will increase current earnings and help meet earnings targets (Roychowdhury, 2006).

Executives use these manipulation tactics to reach immediate earnings benchmarks, usually in the current period. However, altering the very economic activities of the firm may have a detrimental effect on the firm's future performance and may also adversely affect future cash flows (Vorst, 2016). For example, since current R&D expenses are strongly and positively correlated with future performance (e.g., Lev and Sougiannis, 1996), any decline in these expenses will cause a firm to sacrifice growth opportunities that may bear future benefits. Similarly, several studies document that a reduction in advertising expenses negatively affects future revenues (e.g., Cohen et al., 2010). There is also evidence that offering deep sales discounts could be difficult to subdue and may induce a discount war by competitors, eventually decreasing earnings over the long term (Chapman, 2011).

Multiple studies present findings consistent with REM adversely affecting future performance. Eldenburg et al. (2011) report that some non-profit hospitals manipulate discretionary spending and change the timing of the disposal of fixed assets to achieve earnings targets. This manipulation eventually negatively impacts net income in the future. Cohen and Zarowin (2010) also find that firms have lower future operating performance when they use REM to manipulate earnings around seasoned equity offerings. Additionally, firms can manage current cash flow levels by manipulating advertising and SG&A expenses, capital expenditures, production, R&D expenses, sales terms, or deferring the recognition of an impairment loss. Such manipulation, however, negatively affects future performance (Filip et al., 2015). In the same vein, Vorst (2016) investigates the reversal effects of an abnormal decrease in discretionary investments on operating performance over the long term. The authors document that this reversal shows REM and that reversing these reductions in abnormal investment may lead to a lower operating performance in the future.

Taken together, prior studies suggest that using REM to manipulate earnings adversely affects future performance. It follows that investors would be interested in knowing whether companies engage in such operational manipulation of earnings and whether REM impacts investor perception. Gunny (2010) examines the repercussions of REM and investors' views of this phenomenon. By investigating stock price movements of firms using REM, Gunny (2010) finds that REM is correlated with vastly lower future cash flows and earnings, and that investors can see through the implications of REM on future cash flows and earnings.

In this study, we investigate the effect of REM on investors' perceptions as informed by financial intermediaries. Specifically, we focus on whether financial analysts detect REM and whether they recognise any impact from it on firms' performance.

Analysts play a significant role in collecting, processing, and disseminating information to institutional and retail investors. Both investors rely heavily on the information analysts provide and value it greatly. Every year, investors spend vast sums of money to purchase recommendations and forecast data from Fist Call, IBES, Zacks, and other vendors. It is, therefore, interesting to examine whether analysts can help investors recognise the implications of REM on firms' performance.

Lee and Chung (2019) investigate whether security analysts respond to firms' REM and find evidence that analysts lower their earnings forecasts and downgrade their stock recommendations for firms with a high level of REM. More recently, Eiler et al. (2021) examine the effects of REM on the properties of analysts' forecasts and find evidence suggesting that analyst forecasts accuracy decreases and their dispersion increases in REM firms. As noted earlier, both studies only investigate the effect of REM on the analysts' short-term horizon forecasts, omitting the long-term horizon forecasts. While the two studies find an association between REM and short-term analyst forecasts, it remains unclear how it may affect long-term analyst forecasts.

Long-term forecasts are relevant to investors incrementally to short-term forecasts. Copeland et al. (2004) show that revisions in long-term forecasts have a more considerable impact on stock prices than those in short-term forecasts. In addition, Di Martino (2019) provides evidence that investors react to long-term growth targets. Jung et al. (2012) also document a more vigorous market response to stock recommendation revisions when financial analysts issue accompanying long-term growth forecasts. Management forecasts regarding long-term earnings growth are also

informative to investors, and analysts incorporate management long-term earnings guidance in their projections (Faurel et al., 2018; Call et al., 2021[1]). On another note, long-term forecasts are more sensitive to changes in a firm's operating environment and disclosures related to operating plans (Srinidhi et al., 2009), thus making them more susceptible to REM.

Prior research seems to support the notion that there may be an association between REM and analyst long-term forecasts. If analysts are able to uncover the manipulation of earnings using real activities, their long-term projections of earnings will reflect this knowledge, implying that they predict a detrimental effect of REM on future performance for firms engaging in REM. This exposition and these arguments lead us to our first hypothesis, stated in its null form:

Hypothesis 1 (H1): Analysts' forecasts regarding long-term earnings growth are not associated with aggressive REM.

Extant research on financial analysts' performance is mixed on whether analysts are able to recognise and process complex information and fully incorporate all available information into their forecasts (Eiler et al., 2021). We argue that the analysts' task becomes more complicated when issuing long-term estimates, and more so in the presence of REM, for two reasons: First, forecasting long-term growth is more complex and requires more time, effort, and resources (Jung et al., 2012). REM adds a layer of complexity to the information environment for analysts, thus making it even harder to detect. When firms use real activities to manipulate earnings, an analyst requires more time and effort to recognise this manipulation effectively and issue long-term growth predictions. Second, forecasting over the long run makes accurate financial predictions even harder for analysts, since the effect of REM may either reverse shortly after the manipulation or even dissipate over time. Since the additional financial analysis challenge caused by REM activities affects long-term growth forecasts, one can posit that REM may also have implications for the accuracy of these forecasts. We, therefore, present our second hypothesis in its null form:

Hypothesis 2 (H2): Long-term earnings growth forecast bias from analysts is not associated with aggressive REM.

3 Variables, research model, and sample

3.1 REM measures

Following prior studies (e.g., Cohen and Zarowin, 2010; Kim and Park, 2014; Lee and Chung, 2019), we devise the following measures for REM: (a.) abnormal levels of cash flow from operations *CFO*, *AB_CFO*, (b.) abnormal production costs *AB_PROD*, and (c.) abnormal research and development costs *AB_RD*. Managers may inflate actual earnings in different ways. First, they may quickly increase sales using aggressive pricing strategies and more lenient sales terms and discounts, resulting in atypically low operational cash *CFO*. Second, they may reduce the cost of sales via an increase in production to divide fixed production costs over a greater unit number, which would

result in an unusually high cost of production *PROD*. Lastly, they can decrease discretionary expenses such as R&D expenses. Thus, abnormally high *AB_PROD* and abnormally low *AB_CFO* and *AB_RD* are consistent with income-increasing REM. To ease the interpretation of the results, we set up all measures *AB_CFO*, *AB_PROD*, and *AB_RD* to be increasing in income.

Following prior studies (e.g., Roychowdhury, 2006; Cohen and Zarowin, 2010; Kim and Park, 2014; Lee and Chung, 2019), we express normal CFO as a linear function of sales and change in sales from the preceding year, as follows:

$$\mathrm{CFO}_t/\mathrm{TA}_{t-1} = \alpha_0 + \alpha_1(1/\mathrm{TA}_{t-1}) + \alpha_2\,(\mathrm{SALES}_t/\mathrm{TA}_{t-1}) + \alpha_3\,(\Delta\mathrm{SALES}_t/\mathrm{TA}_{t-1}) + \varepsilon_t \quad (1)$$

CFO represents the cash flow from operations, *SALES* represents total sales, and *TA* is the Total Assets (See Appendix 1 for operational definitions of the variables). We estimate this model by running a cross-sectional regression for each industry and year. We compute abnormal CFO, *AB_CFO*, as the difference between the actual and predicted values from equation (1).

As discussed earlier, another avenue for REM is manipulating R&D activities to reduce the R&D expense. In line with prior literature (e.g., Roychowdhury, 2006; Zang, 2012; Chan et al., 2015), we use the following model to estimate the expected level of R&D expenditures:

$$\mathrm{RD}_t/\mathrm{AT}_{t-1} = \alpha_0 + \alpha_1(1/\mathrm{TA}_{t-1}) + \alpha_2\,(\mathrm{SALES}_t/\mathrm{TA}_{t-1}) + \alpha_3\,(\Delta\mathrm{SALES}_t/\mathrm{TA}_{t-1}) + \varepsilon_t \quad (2)$$

RD represents R&D expenditure, and the other variables are as defined above (See Appendix 1 for more details). We estimate this model by running a cross-sectional regression for each 2-digit SIC code industry and year. We compute abnormal RD, *AB_RD*, as the difference between the actual and predicted values from equation (2).

In terms of production costs, and following Roychowdhury (2006) and Lee and Chung (2019), we use the following model to estimate the normal level of production cost:

$$\mathrm{PROD}_t/\mathrm{TA}_{t-1} = \alpha_0 + \alpha_1(1/\mathrm{TA}_{t-1}) + \alpha_2\,(\mathrm{SALES}_t/\mathrm{TA}_{t-1}) + \alpha_3\,(\Delta\mathrm{SALES}_t/\mathrm{TA}_{t-1}) + \alpha_4\,(\Delta\mathrm{SALES}_{t-1}/\mathrm{TA}_{t-1}) + \varepsilon_t \quad (3)$$

PROD represents the production costs, defined as the sum of the cost of goods sold and the change in inventory from the previous year. The other variables are as described previously (See Appendix 1 for more details).

To assess models 1 to 3 above, we run cross-sectional regressions by year and industry, subject to the availability of at least fifteen observations. We define industries according to the Fama and French (1997) industry classification. We calculate Abnormal CFO *AB_CFO*, Abnormal R&D *AB_RD*, and abnormal production *AB_PROD* as the residuals from models 1 to 3 above, respectively. We use these operational variables as our proxies for REM activities. Positive values of abnormal production indicate income-increasing REM, whereas positive values of abnormal CFO and abnormal RD are consistent with income-decreasing REM. To facilitate the interpretation of the results, we multiply *AB_CFO* and *AB_RD* by minus one. Therefore, higher values for these variables are consistent with income-increasing REM.

3.2 Research models

To assess the first hypothesis (*H1*), we estimate the following model:

$$\text{FORECAST} = \alpha_0 + \alpha_1 \text{ REMVAR} + \alpha_2 \text{ PAST} + \alpha_3 \text{ STABILITY} + \alpha_4 \text{ EPS} + \alpha_5 \text{ COV} + \alpha_6 \text{ ROA} + \alpha_7 \text{ EP} + \alpha_8 \text{ BTM} + \alpha_9 \text{ TA} + \alpha_{10} \text{ LEV} + \varepsilon_{i,t} \quad (4)$$

Our dependent variable *FORECAST* is the long-term EPS growth forecasted by financial analysts, as reported in I/B/E/S. This forecast covers a period of five years. The main test variable, *REMVAR*, is our generic REM variable. We replace *REMVAR* with one of the three REM measures presented above for each model. Our first research hypothesis examines the relationship between REM and analyst long-term growth forecast. Therefore, we expect statistically significant coefficients on the REM variables in all three models.

To examine hypothesis 2 (H2), we use the following empirical model:

$$\text{FE} = \alpha_0 + \alpha_1 \text{ REMVAR} + \alpha_2 \text{ PAST} + \alpha_3 \text{ STABILITY} + \alpha_4 \text{ EPS} + \alpha_5 \text{ COV} + \alpha_6 \text{ ROA} + \alpha_7 \text{ EP} + \alpha_8 \text{ BTM} + \alpha_9 \text{ TA} + \alpha_{10} \text{ LEV} + \varepsilon_{i,t} \quad (5)$$

The variable *FE* is the dependent variable and represents analysts' long-term earnings growth forecast error. We calculate *FE* as the realised long-term earnings growth minus the long-term earnings growth forecast. Following prior studies (e.g., Dechow et al., 2000; Lee et al., 2008; Esplin, 2022), we use the realised long-term earnings growth calculated by fitting a least squares growth line through the logarithms of the six annual EPS observations, beginning with year t through year $t + 5$. The long-term earnings growth forecast is the variable *FORECAST* defined above.

We include several variables to control for other factors documented in prior literature to affect long-term analyst forecasts and forecast errors. Following Lee et al. (2008), we add the variables *PAST* and *STABILITY*. We add *PAST* to control for actual past long-term EPS growth. *STABILITY* represents earnings stability and controls for the level of *EPS* growth consistency over the past five years, as reported in the I/B/E/S database. We also include the most recent annual earnings per share *EPS*. We follow Chan et al. (2003) and include the variable *COV* to control for the company's information environment effects. Similarly, we add the return on assets *ROA*, the earnings-to-price ratio *EP*, and the book-to-market ratio *BTM* to control for the company's intrinsic performance and growth. We also control for size using the variable *TA* and for leverage using the variable *LEV*. We winsorise all continuous variables by year at the 1 and 99 percentiles. All estimation models include *Year* and *Industry* fixed effects.

3.3 Sample and data

We compute our dependent variables *FORECAST* and *FE* from long-term earnings growth forecast data and actual earnings growth data collected from the Summary History database in I/B/E/S. Our sample spans the years 2002 to 2017. We start with 2002 because prior studies (e.g., Cohen et al., 2008) document that REM activities begin to increase after the passage of SOX. We only keep companies with December fiscal year-ends, as done in similar studies (e.g., Lee et al., 2008; Wong and Zhang, 2014).

For each firm in every year, we only consider growth forecasts issued in April since these forecasts are the same for the following months and since April forecasts already impound all relevant information contained in the financial statements from the year preceding the forecast year.

Following Lee et al. (2008), we eliminate loss firm observations. In addition, since firms operating in regulated and financial industries (SIC codes 6000-6999 and 4400-4999) are subject to different regulations, we exclude them. We then calculate the REM and the control variables utilising financial statements data from Compustat. We collect the stock price and returns data from the Center for Research in Security Prices (CRSP). We then merge the forecast error data with the control variables and REM data. Appendix 2 illustrates the sample selection procedure.

4 Results

4.1 Univariate analyses

In Table 1, we present our descriptive statistics of the long-term EPS growth, the growth forecasts, the forecast errors, the control variables, and the REM measures.

Table 1 Descriptive statistics

Panel A: Sample description								
Variable	*N*	*Mean*	*St. Dev*	*Min*	*Q1*	*Median*	*Q3*	*Max*
FORECAST	14823	13.969	7.512	1.050	9.150	12.500	17.400	41.670
REALISED	14823	7.290	21.625	–46.580	–4.260	7.100	17.720	81.420
FE_MEAN	14823	–6.657	22.511	–65.920	–18.110	–5.460	4.220	66.950
AB_CFO	13922	–0.084	0.129	–0.493	–0.150	–0.053	–0.001	0.202
AB_RD	6075	0.007	0.078	–0.246	–0.011	0.002	0.025	0.337
AB_PROD	14417	–0.041	0.254	–0.933	–0.125	–0.014	0.037	0.939
AB_COMB	6040	–0.212	0.391	–1.673	–0.429	–0.179	0.009	1.186
PAST	14823	13.278	27.102	–45.560	–1.160	10.490	23.230	128.160
STABILITY	14823	41.639	42.995	1.880	12.970	27.210	53.740	236.990
EPS	14823	1.448	1.405	–1.560	0.550	1.143	1.980	6.940
COV	14823	1.446	0.621	0.693	0.693	1.386	1.946	3.045
ROA	14823	0.052	0.071	–0.190	0.013	0.043	0.087	0.273
EP	14823	11.199	32.463	–48.497	0.737	2.412	8.039	195.427
BTM	14823	0.535	0.362	–0.013	0.289	0.465	0.689	1.991
TA	14823	7.760	1.883	3.566	6.410	7.666	8.997	12.530
LEV	14598	0.742	1.047	0.000	0.109	0.440	0.945	6.611

This table provides descriptive statistics for our full sample. We provide the detailed variable definitions in Appendix and winsorise at the 1 and 99 percentile levels all continuous variables. See Appendix 1 for variable definitions.

Table 1 Descriptive statistics (continued)

Panel B: Yearly descriptive statistics					
		Yearly Long-Term EPS Growth			
Year	*N*	*REALISED*	*FORECAST*	*FE*	*PAST*
2002	620	4.231	15.842	–11.597	22.912
2003	738	4.461	16.614	–12.038	22.843
2004	784	6.661	15.867	–9.171	17.190
2005	824	12.670	16.237	–3.508	14.637
2006	815	17.910	16.083	1.798†	14.740
2007	846	16.696	15.166	1.670†	10.725
2008	915	10.596	14.072	–3.447	9.003
2009	973	1.799	13.530	–11.834	8.414
2010	960	–1.347	13.322	–14.772	12.044
2011	969	0.848	13.891	–13.171	17.690
2012	1056	4.724	13.732	–9.136	22.638
2013	1069	12.347	13.793	–1.479†	20.421
2014	1049	9.888	11.205	–1.084†	14.635
2015	1048	6.649	11.753	–4.991	5.374
2016	1090	5.577	12.753	–7.168	2.571
2017	1067	4.858	12.860	–7.850	4.197

Notes: '†' indicates that at the 1% level, the variable is not significantly different from zero. However, there is significance at this level for all other variables using a two-tailed test. See Appendix 1 for variable definitions.

Panel C: Descriptive Statistics by Industry					
		Long-Term EPS Growth by Industry			
Industry	*N*	*REALISED*	*FORECAST*	*FE*	*PAST*
Agriculture, Forestry, Fishing	21	4.221	9.140	–4.918	–5.166
Mining	739	6.547	16.172	–9.408	19.309
Construction	193	10.619	13.010	–2.662	22.429
Manufacturing	5479	7.665	15.175	–7.475	13.757
Transportation and Public Utilities	1903	5.478	10.467	–4.902	9.418
Wholesale Trade	343	8.622	14.802	–6.262	14.302
Retail Trade	485	8.359	17.497	–9.227	16.559
Finance, Insurance, Real Estate	3734	6.170	10.982	–4.832	9.035
Service	1888	9.794	18.148	–8.343	20.142
Public Administration	38	3.311	13.582	–10.612	8.666

See Appendix 1 for variable definitions.

In Panel A of Table 1, the mean (median) value of forecasted as well as realised long-term earnings growth is 13.969 (12.500) and 7.290 (7.100), respectively, which is

comparable to the values reported in Lee et al. (2008). The mean (median) value of the forecast errors *FE* is –6.657 (–5.460), which is in line with previous research findings that long-term earnings projections from analysts are more likely to be positive (Barron et al., 2013). On average, about 4.231 analysts follow a company's earnings. Regarding the REM measures, the mean (median) value of abnormal cash flows from operating activities is –0.084 (–0.053), the mean (median) abnormal production cost measures –0.041 (–0.014), while the mean (median) value of abnormal R&D expense is 0.007 (0.002). The mean (median) value of the aggregate REM measure is –0.212 (–0.179), which is higher compared to the values reported by Eiler et al. (2021). This value indicates that firms may have more REM activities in our sample than in theirs. The mean (median) value of *PAST* and *STABILITY* is 13.278 (10.490) and 41.639 (27.210), which is consistent with Barron et al. (2013). The other variables' mean and median values align with previous research.

In Panel B of Table 1, we present descriptive statistics for the realised long-term EPS growth forecast and forecast error by year. All the realised long-term EPS growth values are positive except in 2010. This exception may be attributed to the effect of the 2009 financial crisis. All long-term growth forecasts are positive. The forecast errors are negative in most years except 2013 and 2014, which indicates that analysts overall issued optimistic long-term growth forecasts over the sample years. The forecast errors in both 2006 and 2007 were positive and not significant.

Panel C of Table 1 presents the descriptive statistics of long-term growth and forecast error by industry. Most of the sample companies are in the manufacturing, finance, insurance, and real estate industries, which is 36.96%, and 25.19%, respectively. Around 12.84% of companies are in the transportation and public utility industries, whereas 12.74% are in the service industry. Forecast errors for all industries are significantly negative, indicating that analysts are optimistic regarding firms' long-term growth across industries.

Table 2 presents the Spearman and Pearson correlation for all the variables included in the models. The table shows that the variable *FORECAST* is negatively related to all three REM measures. The Pearson correlation between *FORECAST* and *AB_CFO* is –0.267 with a p-value < 0.001, and the Pearson correlation between the forecast error variable *FE* and *AB_CFO* is 0.069, with a p-value < 0.001, indicating that the forecasted long-term EPS growth decreases as abnormal operating cash flows increase. The Pearson correlation between *FORECAST* and *AB_RD* is –0.053, with a p-value < 0.001. The Pearson correlation between the forecast error variable *FE* and *AB_RD* is 0.015, which suggests that the forecasted long-term EPS growth decreases as abnormal R&D expense increases. Similarly, the Pearson correlation between *FORECAST* and *AB_PROD* = –0.137, with a p-value < 0.001. The Pearson correlation between the forecast error FE and *AB_PROD* = 0.039, with a p-value < 0.001.

Collectively, the negative (positive) correlations between *FORECAST* (*FE*) and the three REM measures suggest analysts are more pessimistic when REM increases. The correlations between the control variables with each of the long-term EPS growth forecast *FORECAST* and the forecast error *FE* are similar to those in prior studies. The correlation matrix also suggests low correlations among control variables. However, this does not affect our findings.

Table 2 Correlation matrix

Variable	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*
1- FORECAST		**0.086**	**–0.260**	**–0.267**	**–0.053**	**–0.137**	**0.265**	**0.051**	**–0.262**	**0.093**	**0.118**	**–0.146**	**–0.244**	**–0.411**	**–0.185**
2- REALISED	**0.103**		**0.933**	**–0.022**	–0.001	–0.008	**–0.033**	**0.056**	**–0.183**	**0.036**	**–0.059**	**–0.079**	**0.067**	**–0.070**	–0.012
3- FE	**–0.239**	**0.911**		**0.069**	0.015	**0.039**	**–0.120**	**0.038**	**–0.089**	0.002	**–0.098**	**–0.028**	**0.149**	**0.069**	**0.048**
4- AB_CFO	**–0.274**	**–0.034**	**0.070**		**–0.038**	**0.360**	**–0.241**	**0.100**	0.010	**–0.187**	**–0.402**	**–0.105**	**0.339**	**0.115**	**0.162**
5- AB_RD	**–0.046**	0.012	0.021	**–0.031**		0.011	**–0.028**	–0.013	–0.004	**–0.054**	0.003	–0.019	0.010	0.004	0.020
6- AB_PROD	**–0.174**	–0.012	**0.052**	**0.416**	**0.097**		**–0.101**	**0.047**	0.009	**–0.101**	**–0.243**	**–0.057**	**0.190**	**0.084**	**0.107**
7- PAST	**0.269**	**–0.026**	**–0.121**	**–0.237**	**–0.022**	**–0.134**		**–0.096**	**0.133**	**0.109**	**0.332**	**0.019**	**–0.257**	**–0.128**	**–0.106**
8- STABILITY	**0.082**	**0.055**	**0.025**	**0.113**	–0.021	**0.081**	**–0.154**		**–0.159**	**–0.145**	**–0.131**	**–0.088**	**0.169**	**–0.107**	**0.038**
9- EPS	**–0.330**	**–0.214**	**–0.084**	0.013	0.004	0.009	**0.173**	**–0.284**		**0.111**	**0.249**	**0.325**	**–0.071**	**0.453**	0.002
10- COV	**0.149**	**0.055**	0.008	**–0.168**	**–0.050**	**–0.110**	**0.143**	**–0.184**	**0.159**		**0.147**	**0.097**	**–0.273**	**0.279**	**–0.044**
11- ROA	**0.193**	**–0.042**	**–0.107**	**–0.440**	–0.011	**–0.300**	**0.400**	**–0.134**	**0.275**	**0.171**		**0.231**	**–0.412**	**–0.127**	**–0.212**
12- EP	**–0.259**	**–0.081**	**0.014**	**–0.073**	0.020	**–0.056**	**0.102**	**–0.189**	**0.538**	**0.282**	**0.416**		**–0.049**	**0.521**	**0.014**
13- BTM	**–0.315**	**0.016**	**0.130**	**0.405**	**0.041**	**0.290**	**–0.295**	**0.193**	**–0.032**	**–0.285**	**–0.496**	**–0.093**		**0.100**	–0.007
14- TA	**–0.441**	**–0.061**	**0.091**	**0.148**	**0.023**	**0.121**	**–0.111**	**–0.176**	**0.485**	**0.267**	**–0.185**	**0.662**	**0.110**		**0.233**
15- LEV	**–0.344**	**–0.048**	**0.073**	**0.257**	**0.035**	**0.202**	**–0.164**	0.008	**0.143**	–0.004	**–0.326**	**0.142**	**0.108**	**0.381**	

This table provides the correlation for all variables. Pearson (Spearman) correlation coefficients at the bottom (top) diagonal. The bolded numbers are significant at the 5% level or less. See Appendix 1 for variable definitions.

Table 3 Test results for the EPS growth *Forecast* model

		Model 1			*Model 2*			*Model 3*		
		AB_CFO			*AB_PROD*			*AB_RD*		
Variable	*Pred.*	*Coef.*	*t-Stat.*	*p-value*	*Coef.*	*t-Stat.*	*p-value*	*Coef.*	*t-Stat.*	*p-value*
Intercept		26.593	13.900	<0.0001	26.346	13.420	<0.0001	31.517	24.320	<0.0001
AB_CFO	?	–6.052	–7.170	<0.0001	–	–	–	–	–	–
AB_PROD	?	–	–	–	–1.697	–5.980	<0.0001	–	–	–
AB_RD	?	–	–	–	–	–	–	–2.431	–2.270	0.0233
PAST	+	0.047	12.990	<0.0001	0.049	13.160	<0.0001	0.050	9.120	<0.0001
STABILITY	+	0.009	4.390	<0.0001	0.010	4.500	<0.0001	0.013	4.080	<0.0001
EPS	–	–0.718	–9.780	<0.0001	–0.746	–10.260	<0.0001	–0.917	–7.530	<0.0001
COV	+	0.921	5.620	<0.0001	1.091	6.840	<0.0001	1.033	4.080	<0.0001
ROA	+	–11.432	–7.350	<0.0001	–9.142	–5.920	<0.0001	–10.310	–5.540	<0.0001

Table 3 Test results for the EPS growth *Forecast* model (continued)

		Model 1			*Model 2*			*Model 3*		
		AB_CFO			*AB_PROD*			*AB_RD*		
Variable	*Pred.*	*Coef.*	*t-Stat.*	*p-value*	*Coef.*	*t-Stat.*	*p-value*	*Coef.*	*t-Stat.*	*p-value*
EP	+	0.011	2.990	0.0028	0.010	2.820	0.0048	0.011	2.400	0.0165
BTM	–	–3.326	–11.130	<0.0001	–3.249	–10.960	<0.0001	–5.580	–12.550	<0.0001
TA	–	–1.131	–12.790	<0.0001	–1.095	–12.620	<0.0001	–1.525	–11.820	<0.0001
LEV	–	–0.399	–3.860	0.0001	–0.352	–3.430	0.0006	–0.969	-7.000	<0.0001
Fixed Effects (*Year*)			Yes			Yes			Yes	
Fixed Effects (*Industry*)			Yes			Yes			Yes	
N			13,701			4,197			5,976	
Adj. R^2			39.52%			39.23%			43.16%	

This table presents regression results of real earnings management on long-term growth forecast.

Model 1 is specified as: $FORECAST = \alpha 0 + \alpha 1\ AB_CFO + \alpha 2\ PAST + \alpha 3\ STABILITY + \alpha 4\ EPS + \alpha 5\ COV + \alpha 6\ ROA + \alpha 7\ EP + \alpha 8\ BTM + \alpha 9\ TA + \alpha 10\ LEV + \varepsilon_{i,t}$

Model 2 is specified as: $FORECAST = \beta 0 + \beta 1\ AB_PROD + \beta 2\ PAST + \beta 3\ STABILITY + \beta 4\ EPS + \beta 5\ COV + \beta 6\ ROA + \beta 7\ EP + \beta 8\ BTM + \beta 9\ TA + \beta 10\ LEV + \psi i,\ t$

Model 3 is specified as: $FORECAST = \gamma 0 + \gamma 1\ AB_RD + \gamma 2\ PAST + \gamma 3\ STABILITY + \gamma 4\ EPS + \gamma 5\ COV + \gamma 6\ ROA + \gamma 7\ EP + \gamma 8\ BTM + \gamma 9\ TA + \gamma 10\ LEV + \omega i,\ t$

We winsorise at the 1 and 99 percentile all continuous variables by year. Estimation of all models includes *Year* and *Industry* fixed effects. See Appendix 1 for variable definitions.

4.2 Multivariate regression analyses for the long-term EPS growth forecast FORECAST and the REM variables

Table 3 presents multivariate regression results regarding the relationship between the long-term EPS growth forecasts and the three REM measures. The adjusted R^2 for the abnormal operating cash flows model is 39.52%, which suggests that the variables included in the model modestly explain the variation in the long-term EPS growth forecast adequately. The results show that the coefficient on the variable *AB_CFO* is –6.052 (p-value < 0.001), which is significantly negative after we include the past actual long-term growth over the previous five years, the earnings stability, and the other factors that may impact the long-term EPS growth forecast. When *AB_CFO* increases by one unit, the long-term EPS growth forecast decreases by 6.052%. Thus, analysts tend to be more pessimistic and issue lower long-term EPS growth forecasts when a company's *AB_CFO* increases. Results from this analysis allow us to reject our hypothesis H1 in its null form.

The adjusted R^2 for the abnormal production model is 39.23%. The results show that the coefficient on *AB_PROD* is –1.697 (p-value < 0.001), i.e., significantly negative. When *AB_PROD* increases by one unit, the long-term EPS growth forecast decreases by 1.697%. Thus, analysts are more pessimistic and issue lower long-term EPS growth forecasts when a company's *AB_PROD* increases. This result implies that our null hypothesis H1 is also rejected for this specification.

The adjusted R^2 for abnormal R&D expense is 43.16%. We also report a negatively significant coefficient on *AB_RD* ($\gamma 1 = -2.431$ with a p-value < 0.05). A one-unit increase in *AB_RD* is associated with a 2.431% decrease in the long-term EPS growth forecast. Thus, analysts tend to be more pessimistic and issue lower long-term EPS growth forecasts when a company's *AB_RD* increases. Our null hypothesis H1 is, therefore, rejected for the *RD* specification as well. Additionally, the coefficients on the previous actual earnings growth and that on the earnings stability are positive and significant, indicating that these variables are positively associated with the long-term EPS growth forecast. The results of the control variables are as predicted. We also review the variance inflation factor (VIF) for each explanatory variable and find no indication of multicollinearity concerns that would affect our inferences.

Overall, the results from all three model specifications of REM predicate that analysts lower a company's long-term earnings growth forecast when they perceive that the company engages in REM activities. This result is interesting, especially in light of prior evidence that analysts are usually opportunistic and issue optimistic forecasts when they predict long-term earnings growth. In sum, our analyses consistently reject our first null hypothesis that analysts are indifferent to real activities earnings manipulation.

4.3 Multivariate regression analyses for the long-term EPS growth forecast errors FE and the REM variables

Table 4 presents multivariate regression results regarding the relationship between the long-term EPS growth forecast error FE and the three REM measures. The adjusted R^2

for the abnormal operating cash flow specification is 10.69%, suggesting the variables in the model explain some variation in the long-term EPS growth forecast errors. The results show that the coefficient on *AB_CFO* is 6.854 (p-value < 0.01) after controlling for the past long-term earnings growth over the preceding five years, the earnings stability, and the other factors on the long-term EPS growth forecast error variable *FE*. A one-unit increase in *AB_CFO* is associated with a 6.854% increase in the long-term EPS growth forecast error. This result is in accordance with our previous conclusion that analysts are more pessimistic and release lower long-term EPS growth forecasts when a company's *AB_CFO* increases. This result allows us to reject our null hypothesis H2.

The adjusted R^2 for the abnormal production model specification is 10.11%. The coefficient on the variable *AB_PROD* is 0.711 and is not significant. The adjusted R^2 for the abnormal R&D expense model is 9.11%. The results show that the coefficient on *AB_RD* is negative, but not significant. Therefore, among the three REM measures, only the abnormal operating cash flows *AB_CFO* is found to be negatively affecting analysts' long-term EPS growth forecast. This result supports the view that analysts are pessimistic and have a more challenging time forecasting long-term earnings growth accurately when companies' abnormal operating cash flows increase.

4.4 Sensitivity checks

We perform some additional analyses to verify the robustness of our findings. We first compute an aggregate measure of real earnings manipulation by combining the three REM measures *AB_CFO*, *AB_RD*, and *AB_PROD* used above. Prior studies (e.g., Cohen et al., 2008; Cohen and Zarowin, 2010; Zang, 2012; Braam et al., 2015, Choi et al., 2022) use this methodology and variations of it to devise a combined proxy for REM. In this study, we follow Cohen et al. (2008), Braam et al. (2015), and Choi et al. (2022) to calculate our aggregate measure, *AGREM*. We first multiply the abnormal cash flow from operations variable *AB_CFO* and the abnormal research and development expense variable *AB_RD* by –1 so that higher values of these variables indicate the firm is engaged in sales manipulation and cutting research and development expenses. We then add the abnormal cash flows from operations and abnormal research and development multiplied by –1 to the abnormal production costs. Higher levels of this aggregate measure *AGREM* indicate a higher level of REM.

We re-run our main research models as specified in equations 4 and 5 with the long-term EPS growth forecast *FORECAST* and forecast error *FE* variables as dependent variables, respectively, and the aggregate REM measure as the main independent variable. Table 5 presents multivariate regression results from these analyses. The adjusted R^2 for the long-term growth forecast model is 43.93%, which suggests that the variables included in the model explain the variation in the long-term EPS growth forecast acceptably. The result shows that the coefficient on the aggregate REM measure *AGREM* is –2.121, which is significantly negative with a p-value < 0.001. When the aggregate REM measure increases by one unit, the long-term EPS growth forecast decreases by 2.121%.

Table 4 Test results for the forecast error *FE* model

		Model 1			*Model 2*			*Model 3*		
		AB_CFO			*AB_PROD*			*AB_RD*		
Variable	*Pred.*	*Coef.*	*t-Stat.*	*p-value*	*Coef.*	*t-Stat.*	*p-value*	*Coef.*	*t-Stat.*	*p-value*
Intercept		–25.698	–5.440	<0.0001	–24.851	–5.560	<0.0001	–30.860	–7.160	<0.0001
AB_CFO	?	6.854	2.570	0.010	–	–	–	–	–	–
AB_PROD	?	–	–	–	0.711	0.790	0.430	–	–	–
AB_RD	?	–	–	–	–	–	–	–1.202	–0.310	0.755
PAST	–	–0.052	–3.970	<0.0001	–0.059	–4.540	<0.0001	–0.034	–1.890	0.059
STABILITY	+	0.009	1.370	0.172	0.011	1.600	0.111	–0.006	–0.490	0.627
EPS	–	–2.091	–8.980	<0.0001	–2.127	–9.050	<0.0001	–1.816	–4.880	<0.0001
COV	–	1.305	2.510	0.012	0.952	1.880	0.061	0.939	1.200	0.229
ROA	–	13.618	2.460	0.014	11.117	2.040	0.042	12.691	1.790	0.074

Table 4 Test results for the forecast error *FE* model (continued)

		Model 1			*Model 2*			*Model 3*		
		AB_CFO			*AB_PROD*			*AB_RD*		
Variable	*Pred.*	*Coef.*	*t-Stat.*	*p-value*	*Coef.*	*t-Stat.*	*p-value*	*Coef.*	*t-Stat.*	*p-value*
EP	–	–0.030	–3.050	0.002	–0.030	–3.100	0.002	–0.055	–4.040	<0.0001
BTM	+	6.136	6.230	<0.0001	6.362	6.510	<0.0001	8.252	4.740	<0.0001
TA	+	1.334	5.180	<0.0001	1.402	5.420	<0.0001	2.297	5.730	<0.0001
LEV	+	0.394	1.270	0.205	0.225	0.730	0.464	0.783	1.800	0.072
Fixed Effects (*Year*)			Yes			Yes			Yes	
Fixed Effects (*Industry*)			Yes			Yes			Yes	
N			13,701			14,197			5,976	
Adj. R^2			10.69%			10.11%			9.11%	

This table presents regression results of real earnings management on long-term growth forecast error.

Model 1 is specified as: $FE = \alpha 0 + \alpha 1\ AB_CFO + \alpha 2\ PAST + \alpha 3\ STABILITY + \alpha 4\ EPS + \alpha 5\ COV + \alpha 6\ ROA + \alpha 7\ EP + \alpha 8\ BTM + \alpha 9\ TA + \alpha 10\ LEV + \varepsilon_{i,t}$

Model 2 is specified as: $FE = \beta 0 + \beta 1\ AB_PROD + \beta 2\ PAST + \beta 3\ STABILITY + \beta 4\ EPS + \beta 5\ COV + \beta 6\ ROA + \beta 7\ EP + \beta 8\ BTM + \beta 9\ TA + \beta 10\ LEV + \psi i,\ t$

Model 3 is specified as: $FE = \gamma 0 + \gamma 1\ AB_RD + \gamma 2\ PAST + \gamma 3\ STABILITY + \gamma 4\ EPS + \gamma 5\ COV + \gamma 6\ ROA + \gamma 7\ EP + \gamma 8\ BTM + \gamma 9\ TA + \gamma 10\ LEV + \omega i,\ t$

We winsorise at the 1 and 99 percentile all continuous variables by year. Estimation of all models includes *Year* and *Industry* fixed effects. See Appendix 1 for variable definitions.

Table 5 Test results for the aggregate REM measure *AGREM*

Dependent variable		*FORECAST*			*FE*		
Variable	*Pred.*	*Coef.*	*t-Stat.*	*p-value*	*Coef.*	*t-Stat.*	*p-value*
Intercept		31.141	23.630	<0.0001	–30.450	–7.000	<0.0001
AGREM	?	–2.121	–6.860	<0.0001	2.813	2.880	0.004
PAST	–	0.048	8.920	<0.0001	–0.029	–1.640	0.101
STABILITY	+	0.012	4.000	<0.0001	–0.005	–0.400	0.690
EPS	–	–0.856	–7.070	<0.0001	–1.878	–5.060	<0.0001
COV	–	0.832	3.300	0.001	1.277	1.600	0.111
ROA	–	–12.354	–6.530	<0.0001	14.991	2.080	0.038
E/P	–	0.009	2.030	0.043	–0.052	–3.800	0.000
BTM	+	–5.319	–12.010	<0.0001	8.069	4.570	<0.0001
TA	+	–1.485	–11.540	<0.0001	2.225	5.510	<0.0001
LEV	+	–0.919	–6.660	<0.0001	0.704	1.620	0.105
Fixed Effects (Year)			Yes			Yes	
Fixed Effects (Industry)			Yes			Yes	
N			5,941			5,941	
Adj. R^2			43.93%			9.30%	

This table presents regression results of combine real earnings management on long-term growth forecast and forecast error. The Forecast model is specified as: $FORECAST = \alpha 0 + \alpha 1\ AGREM + \alpha 2\ PAST + \alpha 3\ STABILITY + \alpha 4\ EPS + \alpha 5\ COV + \alpha 6\ ROA + \alpha 7\ EP + \alpha 8\ BTM + \alpha 9\ TA + \alpha 10\ LEV + \varepsilon_{i,t}$; whereas the FE model is specified as: $FE = \beta 0 + \beta 1\ AGREM + \beta 2\ PAST + \beta 3\ STABILITY + \beta 4\ EPS + \beta 5\ COV + \beta 6\ ROA + \beta 7\ EP + \beta 8\ BTM + \beta 9\ TA + \beta 10\ LEV + \psi i,\ t$ We winsorise at the 1 and 99 percentile all continuous variables by year. Estimation of all models includes *Year* and *Industry* fixed effects. See Appendix 1 for variable definitions.

The adjusted R^2 for the long-term growth forecast error model is 9.30%, suggesting that the variables in the model explain some of the variation in long-term EPS growth forecast error. The results show that the coefficient on the aggregate REM measure *AGREM* is 2.813 with a p-value < 0.01, which is significantly positive. This coefficient indicates that when the aggregate measure of REM increases by one unit, the long-term EPS growth forecast error increases by 2.813%. Thus, the result is consistent with our previous findings that analysts are more pessimistic and provide a lower long-term EPS growth forecast in the presence of higher REM. We thus reject both our null hypotheses, H1 and H2, for the aggregate measure specifications.

Second, since negative forecast errors indicate analysts are more opportunistic about the company's long-term EPS growth, we divide our sample into a positive long-term EPS growth forecast errors sub-sample and a negative long-term EPS growth forecast errors sub-sample. We use the models with the three REM measures as independent variables and run cross-sectional regressions on the negative forecast errors sub-sample only.

Table 6 Test Results for the negative forecast error sub-sample

		Model 1			*Model 2*			*Model 3*			*Model 4*		
		AB_CFO			*AB_PROD*			*AB_RD*			*AGREM*		
Variable	*Pred.*	*Coef.*	*t-Stat.*	*p-value*	*Coef.*	*t-Stat.*	*p-value*	*Coef.*	*t-Stat.*	*p-value*	*Coef.*	*t-Stat.*	*p-value*
Intercept		–16.761	–8.510	<0.0001	–16.439	–7.960	<0.0001	–19.021	–11.340	<0.0001	–18.718	–11.290	<0.0001
AB_CFO	?	4.377	3.900	0.000	–	–	–	–	–	–	–	–	–
AB_PROD	?	–	–	–	1.315	2.960	0.003	–	–	–	–	–	–
AB_RD	?	–	–	–	–	–	–	1.864	1.020	0.307	–	–	–
AGREM	?	–	–	–	–	–	–	–	–	–	1.783	4.060	<0.0001
PAST	–	–0.037	–7.080	<0.0001	–0.038	–7.230	<0.0001	–0.046	–6.160	<0.0001	–0.044	–6.050	<0.0001
STABILITY	+	–0.011	–4.180	<0.0001	–0.011	–4.230	<0.0001	–0.010	–2.660	0.008	–0.009	–2.440	0.015
EPS	–	0.265	2.720	0.007	0.270	2.850	0.005	0.362	2.470	0.014	0.305	2.060	0.039
COV	–	0.535	2.360	0.018	0.345	1.600	0.111	0.515	1.530	0.126	0.677	2.010	0.045
ROA	–	6.191	2.500	0.013	4.438	1.810	0.070	5.084	1.760	0.079	6.822	2.330	0.020

Table 6 Test Results for the negative forecast error sub-sample (continued)

		Model 1			*Model 2*			*Model 3*			*Model 4*		
		AB_CFO			*AB_PROD*			*AB_RD*			*AGREM*		
Variable	*Pred.*	*Coef.*	*t-Stat.*	*p-value*	*Coef.*	*t-Stat.*	*p-value*	*Coef.*	*t-Stat.*	*p-value*	*Coef.*	*t-Stat.*	*p-value*
EP	–	–0.016	–3.290	0.001	–0.017	–3.440	0.001	–0.022	–3.490	0.001	–0.021	–3.340	0.001
BTM	+	0.941	2.000	0.045	0.666	1.430	0.154	1.859	2.590	0.010	1.493	2.080	0.038
TA	+	0.784	6.860	<0.0001	0.766	6.820	<0.0001	1.054	6.480	<0.0001	1.021	6.370	<0.0001
LEV	+	0.048	0.310	0.755	–0.042	–0.270	0.784	0.506	2.680	0.007	0.434	2.250	0.025
Fixed Effects (*Year*)			Yes			Yes			Yes			Yes	
Fixed Effects (*Industry*)			Yes			Yes			Yes			Yes	
N			8,827			9,231			3,945			3,921	
Adj. R^2			17.15%			17.38%			19.82%			20.39%	

This table presents regression results of real earnings management on negative long-term growth forecast error.

Model 1 is specified as: $FORECAST = \alpha 0 + \alpha 1\ AB_CFO + \alpha 2\ PAST + \alpha 3\ STABILITY + \alpha 4\ EPS + \alpha 5\ COV + \alpha 6\ ROA + \alpha 7\ EP + \alpha 8\ BTM + \alpha 9\ TA + \alpha 10\ LEV + \varepsilon_{i,t}$

Model 2 is specified as: $FORECAST = \beta 0 + \beta 1\ AB_PROD + \beta 2\ PAST + \beta 3\ STABILITY + \beta 4\ EPS + \beta 5\ COV + \beta 6\ ROA + \beta 7\ EP + \beta 8\ BTM + \beta 9\ TA + \beta 10\ LEV + \psi i,\ t$

Model 3 is specified as: $FORECAST = \gamma 0 + \gamma 1\ AB_RD + \gamma 2\ PAST + \gamma 3\ STABILITY + \gamma 4\ EPS + \gamma 5\ COV + \gamma 6\ ROA + \gamma 7\ EP + \gamma 8\ BTM + \gamma 9\ TA + \gamma 10\ LEV + \omega i,\ t$

Model 4 is specified as: $FORECAST = \delta 0 + \delta 1\ AGREM + \delta 2\ PAST + \delta 3\ STABILITY + \delta 4\ EPS + \delta 5\ COV + \delta 6\ ROA + \delta 7\ EP + \delta 8\ BTM + \delta 9\ TA + \delta 10\ LEV + \eta i,\ t$

We winsorise at the 1 and 99 percentile all continuous variables by year. Estimation of all models includes *Year* and *Industry* fixed effects. See Appendix 1 for variable definitions.

Table 6 presents the regression results for the relationship between the negative long-term EPS growth forecast errors and the three REM measures. The adjusted R^2 for abnormal operating cash flows specification is 17.15%. The results also show that the coefficient on *AB_CFO* is 4.377 (p-value < 0.001), which is significantly positive after we control for the other factors.

The adjusted R^2 for the abnormal production specification is 17.38%. The results show that the coefficient on *AB_PROD* is 1.315 (p-value < 0.05), which is significantly positive. The adjusted R^2 for the abnormal R&D expense is 19.82%. The results show that the coefficient of *AB_RD* is positive but not significant after controlling for the other factors that may impact the long-term EPS growth forecast error. The adjusted R^2 for the aggregate REM measure is 20.39%. The results show that the coefficient on the aggregate measure *AGREM* is 1.783 (p-value < 0.001), which is significantly positive. Thus, the additional analyses support the conclusion that analysts are more pessimistic and issue a lower long-term EPS growth forecast when a company's REM increases. The results from this analysis corroborate our earlier findings.

5 Conclusion

This study examines whether analysts recognise earnings management through manipulating real activities. We use three measures of REM — *AB_CFO*, *AB_PROD*, and *AB_RD* – and find strong evidence that analysts lower a company's long-term earnings growth forecast if they perceive that the company engages in REM activities. Moreover, we surmise that analysts are pessimistic when companies manipulate earnings through real activities. These results are robust to sensitivity checks, including an aggregate measure of REM and a sub-sample of negative forecast error observations.

While the results of this research are novel, the study may be tinted with a few limitations inherent in most REM studies. First, this research focuses primarily on a subset of the most prevalent REM tools proposed in prior accounting research. As discussed above, there are other avenues of REM manipulation that we did not consider in the present study. Moreover, REM relies on expense line items presented in the firms' income statements. These expenses may be affected by management accrual choices. For example, Zang (2012) provides evidence that firms may engage in both AEM and REM. In this context, it is complicated to devise variables completely free of measurement errors or manipulations. Another limitation is that analysts may self-select to monitor organisations that do not engage in REM, which presents a potential endogeneity issue for our analyses. Lastly, these REM measures could potentially include measurement errors, which would impact the power of our tests. While recent accounting research addresses this issue (e.g., Srivastava, 2019; Cohen et al., 2020), very few studies use the newly derived measures. Readers and researchers should interpret the results of this study in light of these limitations.

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Note

[1]Call et al. (2021) find conflicting and only marginally significant results about the association between the issuance of long-term earnings guidance and income-increasing real earnings management.

Appendix 1. Variable definitions

FORECAST	Analyst long-term earnings growth forecast reported by analysts in the nearest month after the earnings announcement
REALIZED	Realized long-term earnings growth over five years. I/B/E/S calculates the actual five-year earnings per share growth by fitting a least squares growth line through the logarithms of the six annual EPS observations beginning with year t through year $t + 5$
FE	Forecast error, calculated as the realized earnings growth (REALIZED) minus the forecasted EPS growth (FORECAST), both defined above
CFO	Cash flow from operating activities, which is the Cash flow from operations (OANCF) scaled by total assets at the beginning of the forecast year
RD	R&D Expenditure, which is the R&D expense (XRD) scaled by total assets at the beginning of the forecast year
PROD	Production cost, calculated as the sum of the cost of goods sold and the change in inventory scaled by total assets at the beginning of the year
SALES	Sales is the total sales for the firm in the year preceding the forecast year
AB_CFO	Abnormal cash flow from operations, calculated as the residual value from cash flows from operating activities equation 1
AB_RD	Abnormal R&D expenditures, calculated as the residual value of the research and development expense equation 2
AB_PROD	Abnormal production, calculated as the residual value of the production cost equation 3
AGREM	Aggregate REM measure, calculated as: AGREM = AB_PROD + (–1)×AB_CFO + (–1)×AB_RD
PAST	Past is actual long-term growth over the previous five years at the time analysts issue growth forecasts, as reported in I/B/E/S

STABILITY	Stability of earnings reported in the I/B/E/S database, calculated as the mean absolute difference between actual reported EPS and a trend line of earnings per share growth over the past five years
EPS	EPS is the most recent annual earnings per share before the the analyst issues the growth forecast, as reported in I/B/E/S
COV	Analyst coverage, measured 8 months before a firm's fiscal year-end
ROA	Return on Assets, measured as income before extraordinary items scaled by average total assets in the fiscal year preceding the forecast
EP	Earnings-to-price ratio, calculated as income before extraordinary items divided by the market value of common equity for the year preceding the forecast year
BTM	Book to market ratio, measured as the book value of equity divided by the market value of equity at the end of the year preceding the forecast year
TA	Total assets for the firm at the beginning of the forecast year
LEV	Leverage is long-term debt divided by book value for the firm before the forecast year

Appendix 2. Sample selection procedure

Long-Term Growth Forecast from I/B/E/S (2002–2017)	47,670 observations
Merge with Actual Earnings	(18,534)
Merge with REM	(6061)
Merge with control variables	(9374)
Final Sample	13,701 observations